13.3 BT9100

13.3.1 General information

The bus transmitter BT9100 provides for the seamless expansion of the X20 System. The stations can be up to 100 m away from each other.

- X2X Link bus transmitter
- For seamless expansion of the system
- Up to 100 m segment lengths
- Feed for internal I/O supply

Information:

If the feed is being used for internal I/O supply, this potential group can not be supplied from any other module. An I/O module with bus module BM01 should be used to separate the potential group (see chapter 3 "Mechanical and electrical configuration", section 16.2 "Supply feed via bus transmitter" on page 88).

13.3.2 Order data

Model number	Short description	Image
	Bus transmitter	
X20BT9100	X20 bus transmitter (X2X Link)	
	Required accessories	
X20TB06	X20 terminal block, 6-pin, 24 V coded	
X20TB12	X20 terminal block, 12-pin, 24 V coded	
X20BM11	X20 bus module, 24 V coded, internal I/O supply is interconnected	
	Optional accessories	
X67CA0X99.1000	Cable for custom prefabrication, 100.0 m	

Table 170: BT9100 - order data

13.3.3 Technical data

Product ID	BT9100
Short description	
Bus transmitters	X2X Link bus transmitter with supply for I/O
I/O supply input	
Input voltage	24 VDC (-15% / +20%)
Fuse	Recommended pre-fusing max. 10 A slow-blow
I/O supply output	
Rated output voltage	24 VDC
Permitted contact load	10.0 A
General information	
Status indicators	X2X bus function, operating status, module status
Diagnostics Module run/error X2X bus function	 Yes, with status LED and software status Yes, with status LED
Power consumption 1) Bus I/O internal	 0.2 W 0.6 W
Certification	CE, C-UL-US, GOST-R
Operational conditions	
Operating temperature Horizontal installation Vertical installation	 0°C to +55°C 0°C to +50°C
Relative humidity	5 to 95%, non-condensing
Mounting orientation	Horizontal or vertical
Installation at altitudes above sea level 0 - 2000 m >2000 m	 No derating Reduction of ambient temperature by 0.5°C per 100 m
Protection type	IP20
Storage and transport conditions	
Temperature	-25°C to +70°C
Relative humidity	5 to 95%, non-condensing
Mechanical characteristics	
Spacing	$12.5^{+0.2}$ mm
Comment	Order terminal block 1 x X20TB06 or X20TB12 separately Order bus module 1x X20BM11 separately

Table 171: BT9100 - technical data

1) The specified values are maximum values. The exact calculation is described in chapter 3 "Mechanical and electrical configuration", section 20 "Power consumption of X20 supply and potential distribution modules" on page 104. The calculation is also available for download as a data sheet with the other module documentation on the B&R homepage.

13.3.4 Additional technical data

Product ID	BT9100
I/O supply input	
Reverse polarity protection	No
I/O supply output	
Behavior if a short circuit occurs	Recommended protection
General information	
B&R ID code	$1BC2

Table 172: BT9100 - additional technical data

13.3.5 Status LEDs

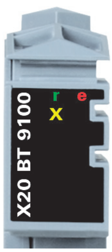
Image	LED	Color	Status	Description
	r	Green	Off	Module supply not connected
			Single flash	Reset mode
			Blinking	Preoperational mode
			On	RUN mode
	e	Red	Off	Module supply not connected or everything is OK
			Double flash	Indicates one of the following conditions: • I/O supply too low • X2X bus supply too low
	e + r	Steady red / single green flash		Invalid firmware
	X	Orange	Off	No communication at the X2X Link
			On	X2X Link communication in progress

Table 173: BT9100 - status indicators

13.3.6 Pin assignments

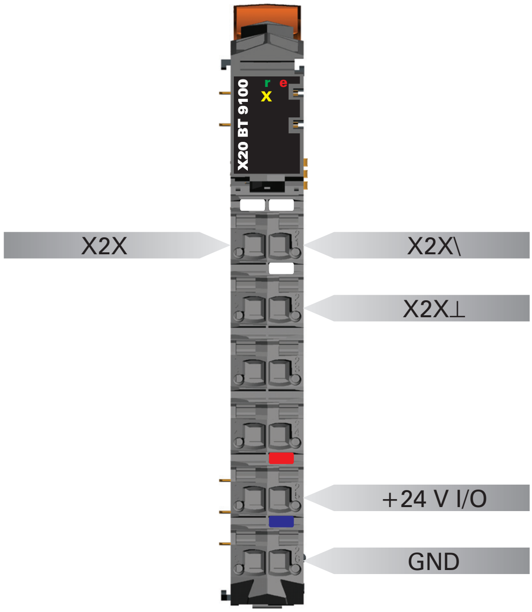


Figure 129: BT9100 - pin assignments

13.3.7 Connection examples

No feed for internal I/O supply

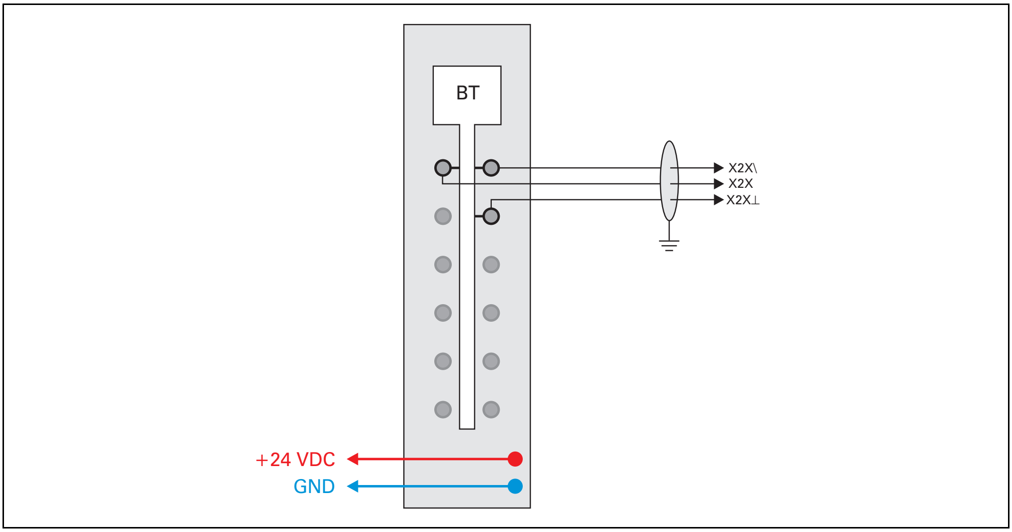


Figure 130: BT9100 - connection example - no feed for internal I/O supply

With feed for internal I/O supply

See also chapter 3 "Mechanical and electrical configuration", section16.2 "Supply feed via bus transmitter" on page 88.

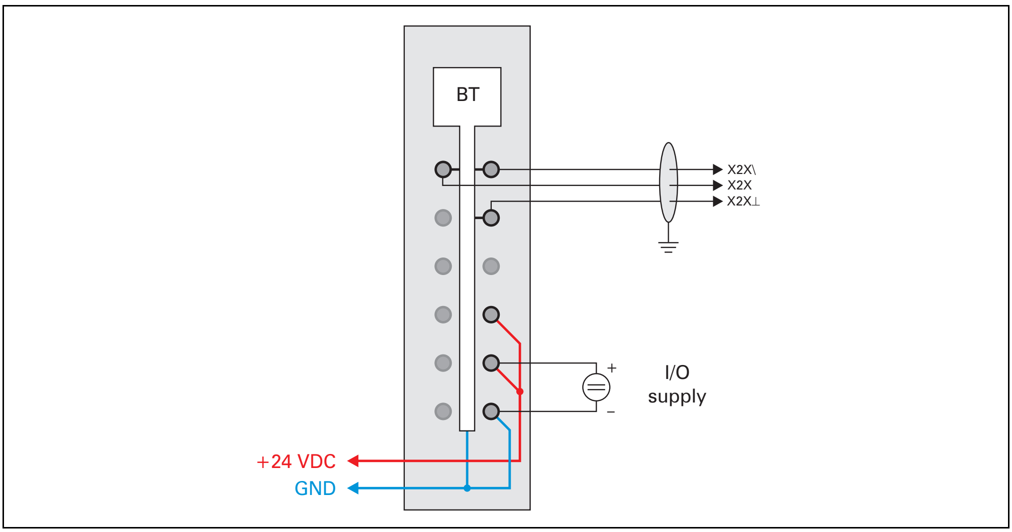


Figure 131: BT9100 - connection example - with feed for internal I/O supply

13.3.8 Connection to next X2X Link I/O node

The bus transmitter BT9100 establishes the connection to the next X2X Link based I/O node. It is important to be sure that only the data lines are connected on. X2X Link supply is system dependant.

System	X2X Link supply
X67 System	System supply PS1300
Remote I/O with X2X Link (XX modules)	24 VDC external supply
Remote valve manifold connection (XV modules)	24 VDC external supply

Table 174: BT9100 - system dependant X2X Link supply

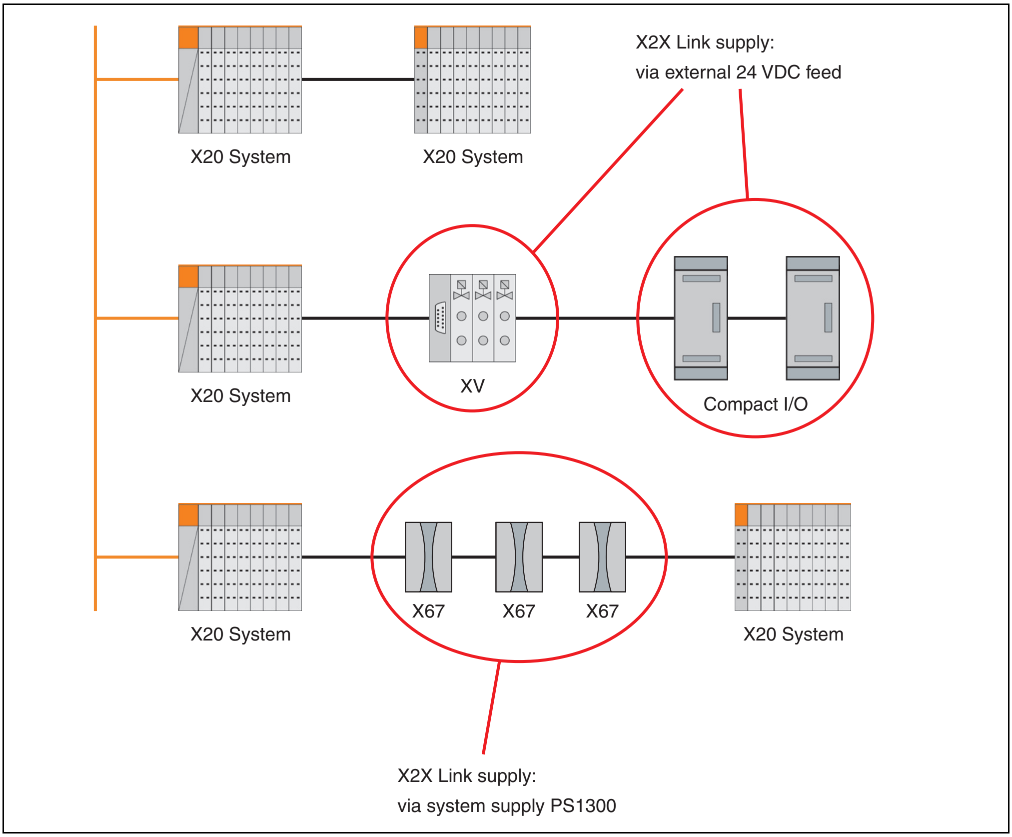


Figure 132: BT9100 - X2X Link supply according to system

13.3.9 Module status

The following module states are monitored:

- Bus supply voltage
- 24 VDC I/O supply voltage

The module provides the status either as a bit or a value. The following table provides an overview:

Module status		Description
Bus supply voltage	Bit	A bus supply voltage of <4.7 V is displayed as a warning.
	Value	The bus supply voltage is measured with a resolution of 0.1 V.
24 VDC I/O supply voltage	Bit	An I/O supply voltage of <20.4 V is displayed as a warning.

13.3.10 B&R ID code

Code for module identification ($1BC2).

13.3.11 Minimum cycle time

The minimum cycle time is the minimum time needed for the bus cycle to be shut down without a communication error occurring. It should be noted that very fast cycles have reduced the idle time needed for handling monitoring, diagnostics and acyclic commands.

Minimum cycle time
≥100 μs

13.3.12 Minimum I/O update time

The minimum I/O update time refers to the minimum time it takes for the bus cycle to shut down, so that in each cycle an I/O update takes place.

Minimum I/O update time
2 ms